

An **Observable is a wrapper around asynchronous data. We use an observable to handle asynchronous data.**



What is synchronous programming

JavaScript is a **single-threaded** programming language.

- The code is executed line by line one after the other in the order in which they are defined.
- Synchronous code is blocking in nature.



What is asynchronous programming

Asynchronous code does not execute in **single-threaded**. It gets executed in the background.

- Asynchronous code is blocking in nature.



Promise vs Observable

- A promise cannot handle stream of asynchronous data. It always returns a single value. On the other hand, we can use observables to handle stream of asynchronous data. It can return multiple values.
- A promise will certainly return a data even if no code is using that data. Whereas an observable will return a data only if someone is going to use that data.
- A promise is native to JavaScript program. Whereas observable is not native to JavaScript and it is provided by “**RxJS**” library.

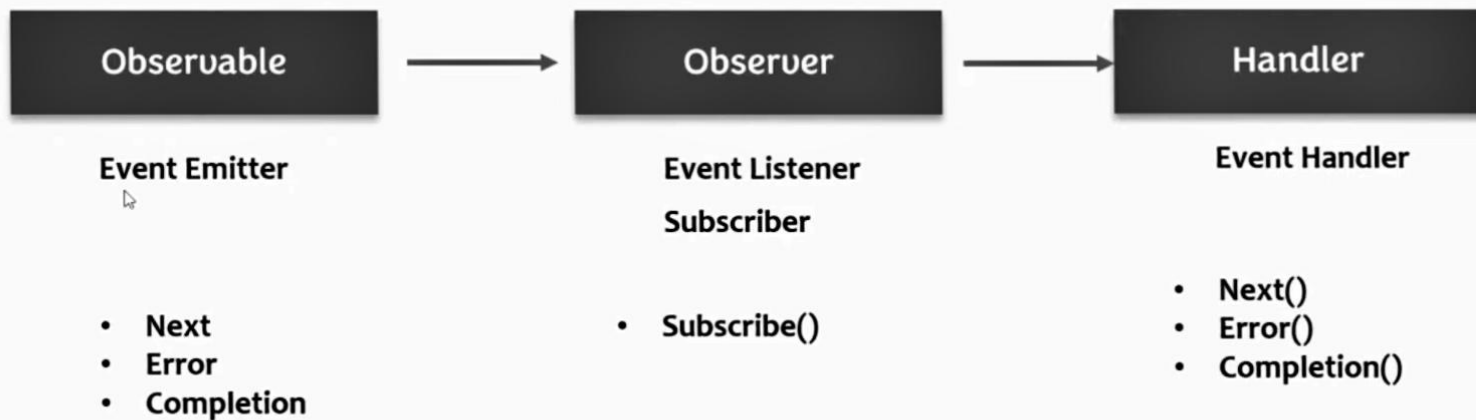


What is an Observable?

An Observable is a function that converts the ordinary data stream into an **observable** one. You can think of Observable as a wrapper around the ordinary data stream.



Observable Pattern



What is an Observable?

An Observable is a function that converts the ordinary data stream into an **observable** one. You can think of Observable as a wrapper around the ordinary data stream.

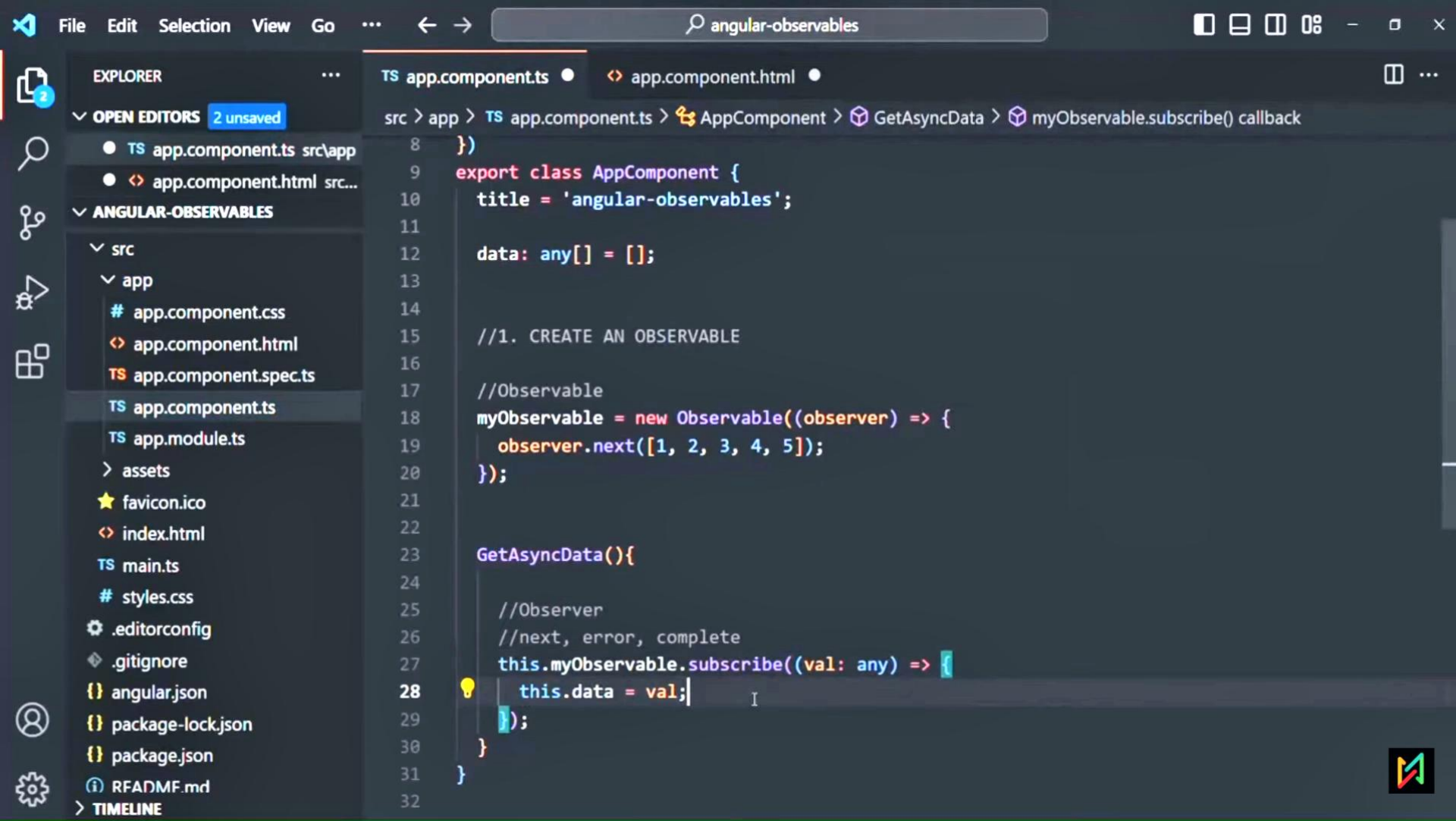


RxJS Main Players



RxJS Main Players





File

Edit

Selection

View

Go

...

←

→

angular-observables

EXPLORES

...

OPEN EDITORS

TS app.component.ts src\app

app.component.html src...

ANGULAR-OBSERVABLES

src

app

app.component.css

app.component.html

app.component.spec.ts

app.component.ts

app.module.ts

assets

favicon.ico

index.html

main.ts

styles.css

.editorconfig

.gitignore

angular.json

package-lock.json

package.json

RFADMF.md

TIMELINE

TS app.component.ts X

app.component.html

src > app > TS app.component.ts > AppComponent > GetAsyncData > myObservable.subscribe() callback

15

//1. CREATE AN OBSERVABLE

16

17

//Observable

18

myObservable = new Observable((observer) => {

19

observer.next(1);

20

observer.next(2);

21

observer.next(3);

22

observer.next(4);

23

observer.next(5);

24

});

25

26

27

GetAsyncData(){

28

29

//Observer

30

//next, error, complete

31

this.myObservable.subscribe((val: any) => {

32

this.data.push(val);

33

});

34

}

35

}

36

File

Edit

Selection

View

Go

...

←

→

angular-observables

EXPLORER

...

OPEN EDITORS1 unsaved

● TS app.component.ts src\app

<> app.component.html src...

ANGULAR-OBSERVABLES

▼ src

▼ app

app.component.css

<> app.component.html

TS app.component.spec.ts

TS app.component.ts

TS app.module.ts

> assets

★ favicon.ico

<> index.html

TS main.ts

styles.css

⚙ .editorconfig

📄 .gitignore

{ } angular.json

{ } package-lock.json

{ } package.json

📄 RFADMF.md

> TIMELINE

TS app.component.ts ●

<> app.component.html

src > app > TS app.component.ts > AppComponent > myObservable > <function> > setTimeout() callback

11

12

13

14

15

16

17

18

19

20

21

22

23

24

25

26

27

28

29

30

31

32

33

34

35

36

data: any[] = [];

//1. CREATE AN OBSERVABLE

//Observable

myObservable = new Observable((observer) => {

setTimeout(() => { observer.next(1) }, 1000);

setTimeout(() => { observer.next(2) }, 2000);

setTimeout(() => { observer.next(3) }, 3000);

setTimeout(() => { observer.error(new Error('Something went wrong. Please try again later!'))

setTimeout(() => { observer.next(4) }, 4000);

setTimeout(() => { observer.next(5) }, 5000);

});

GetAsyncData(){

//Observer

//next, error, complete

this.myObservable.subscribe((val: any) => {

this.data.push(val);

});

}

File

Edit

Selection

View

Go

...

←

→

angular-observables

0%

EXPLORER

...

▼ OPEN EDITORS

✕ TS app.component.ts src\app

🔗 app.component.html src...

▼ ANGULAR-OBSERVABLES

▼ src

▼ app

app.component.css

🔗 app.component.html

TS app.component.spec.ts

TS app.component.ts

TS app.module.ts

> assets

★ favicon.ico

🔗 index.html

TS main.ts

styles.css

⚙ .editorconfig

📄 .gitignore

{ } angular.json

{ } package-lock.json

{ } package.json

📄 RFADMF.md

> TIMELINE

TS app.component.ts

🔗 app.component.html

src > app > TS app.component.ts > AppComponent > myObservable > <function>

11

12

13

14

15

16

17

18

19

20

21

22

23

24

25

26

27

28

29

30

31

32

33

34

35

36

data: any[] = [];

//1. CREATE AN OBSERVABLE

//Observable

myObservable = new Observable((observer) => {

setTimeout(() => { observer.next(1) }, 1000);

setTimeout(() => { observer.next(2) }, 2000);

setTimeout(() => { observer.next(3) }, 3000);

setTimeout(() => { observer.error(new Error('Something went wrong. Please try again later!'))

setTimeout(() => { observer.next(4) }, 4000);

setTimeout(() => { observer.next(5) }, 5000);

});

GetAsyncData(){

//Observer

//next, error, complete

this.myObservable.subscribe((val: any) => {

this.data.push(val);

});

}

Observables in Angular

1

2

3

Get Data



<> app.component.html

```
src > app > TS app.component.ts > AppComponent > myObservable > <function> > setTimeout() callback
```

15

16

16

17

19

10

19

20

21

22

22

23

24

25

25

26

27

28

29

30

31

22

32

33

34

25

33

36

37

38

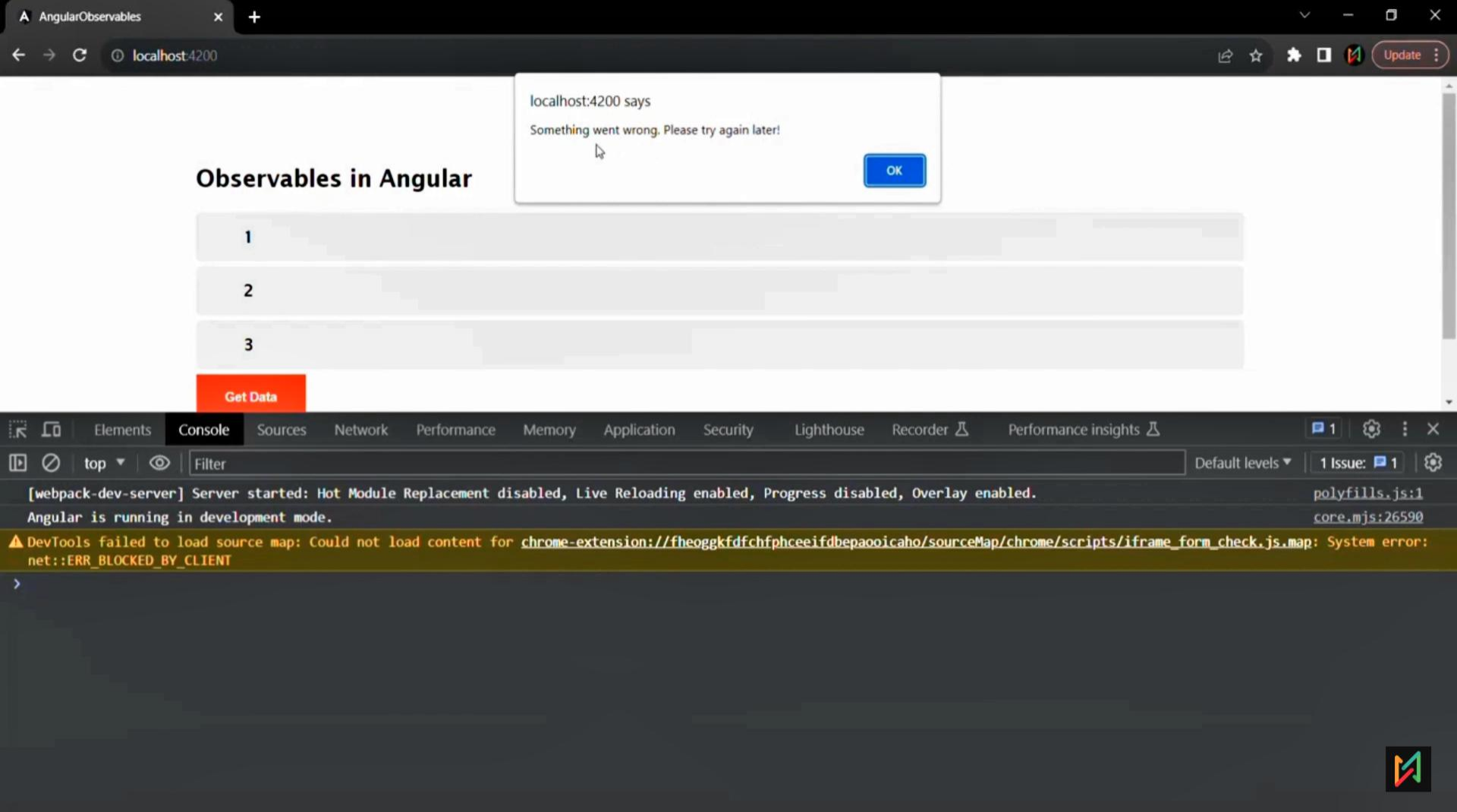
39

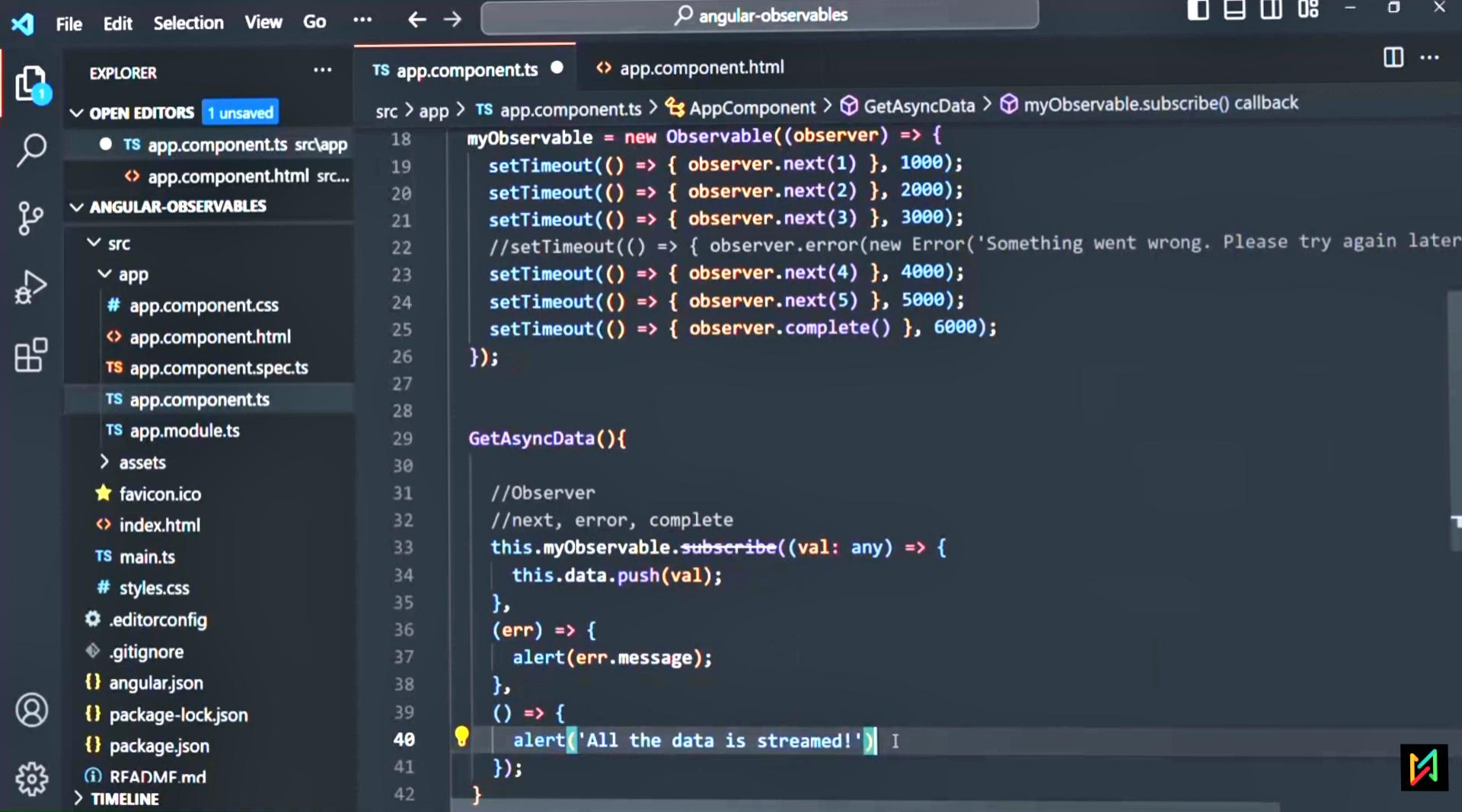
} ,

```
});
```

}







AngularObservables

localhost:4200

Update

1

2

3

4

5

Get Data

localhost:4200 says
All the data is streamed!

OK

Elements

Console

Sources

Network

Performance

Memory

Application

Security

Lighthouse

Recorder

Performance insights

top

Filter

Default levels

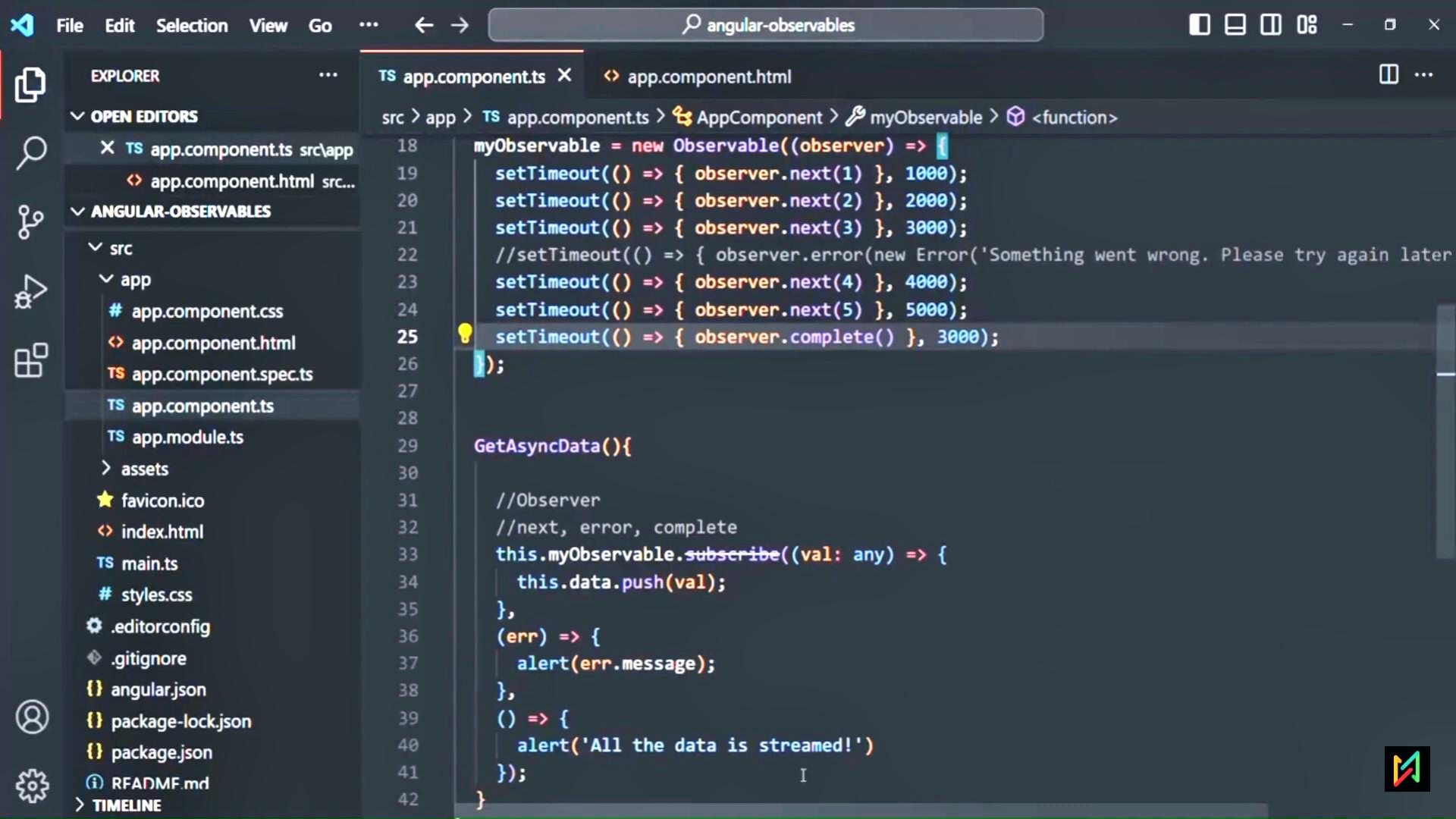
1 Issue

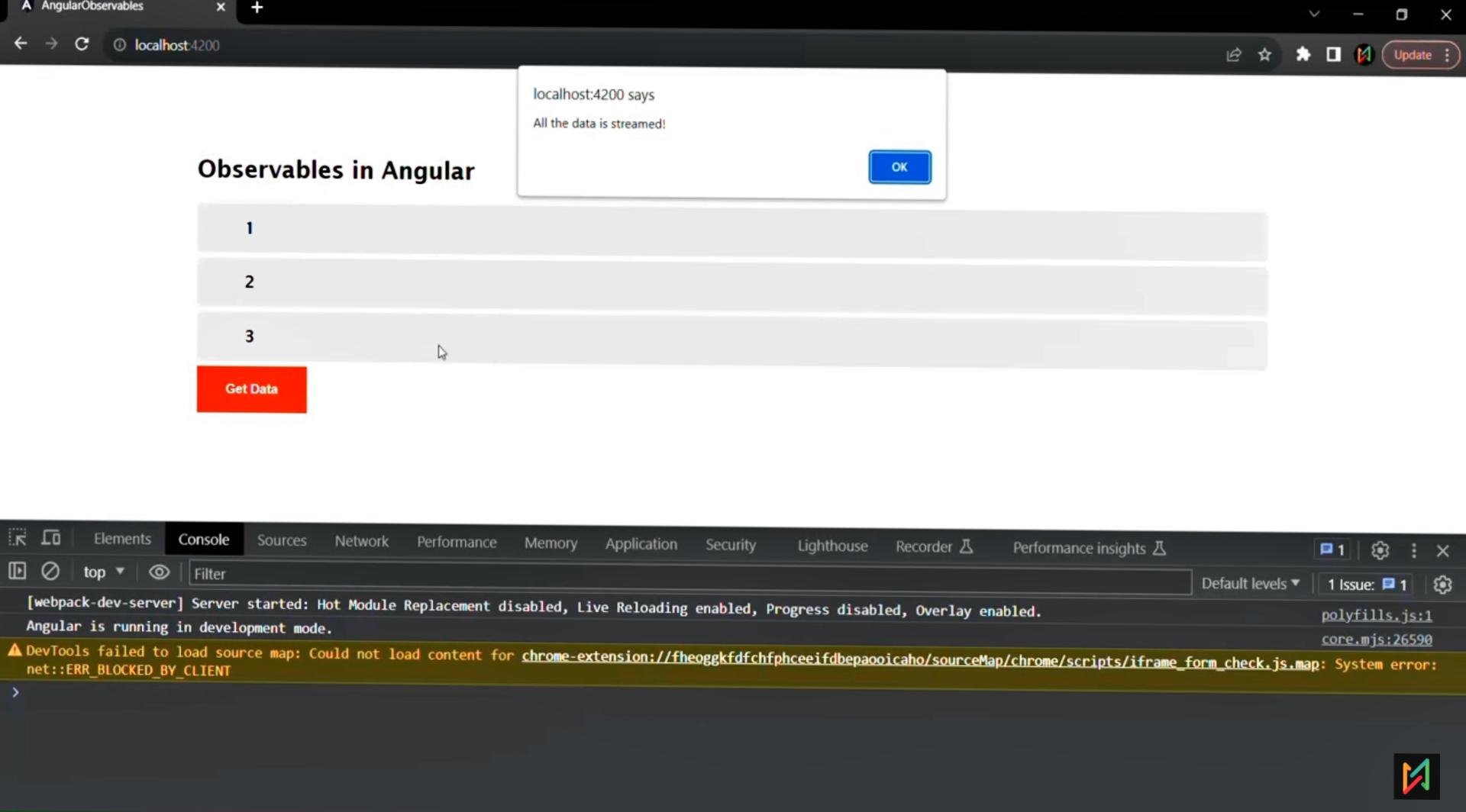
[webpack-dev-server] Server started: Hot Module Replacement disabled, Live Reloading enabled, Progress disabled, Overlay enabled.

Angular is running in development mode.

DevTools failed to load source map: Could not load content for chrome-extension://fheoggkfdfchfphceeiadbepaoaicaho/sourceMap/chrome/scripts/iframe_form_check.js.map: System error: net::ERR_BLOCKED_BY_CLIENT







Observables in Angular

1

2

3

Get Data

localhost:4200 says

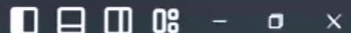
All the data is streamed!

OK

[webpack-dev-server] Server started: Hot Module Replacement disabled, Live Reloading enabled, Progress disabled, Overlay enabled.

Angular is running in development mode.

⚠ DevTools failed to load source map: Could not load content for chrome-extension://fheoggkfdfchfpheeifdbepaooicaho/sourceMap/chrome/scripts/iframe_form_check.js.map: System error: net::ERR_BLOCKED_BY_CLIENT



TS app.component.ts • app.component.html

```
src > app > TS app.component.ts > AppComponent > myObservable > <function>
```

```
18  e = new Observable((observer) => {
```

```
19 t(() => { observer.next(1) }, 1000);
```

```
20 t(() => { observer.next(2) }, 2000);
```

```
21 t(()) => { observer.next(3) }, 3000);
```

```
22 t(() => { observer.error(new Error('Something went wrong. Please try again later!')) }, 3000);
```

```
23 t(() => { observer.next(4) }, 4000);
```

```
24 t(() => { observer.next(5) }, 5000);
```

```
25 t(() => { observer.complete() }, 3000);
```

26

29 $m()$

30

```
32  error, complete
```

```
33 servable.subscribe((val: any) => {
```

```
33     set variavelSubstrado((val1, any)) => {
34         to push(val1);
```

```
34 ta.push(val);
```

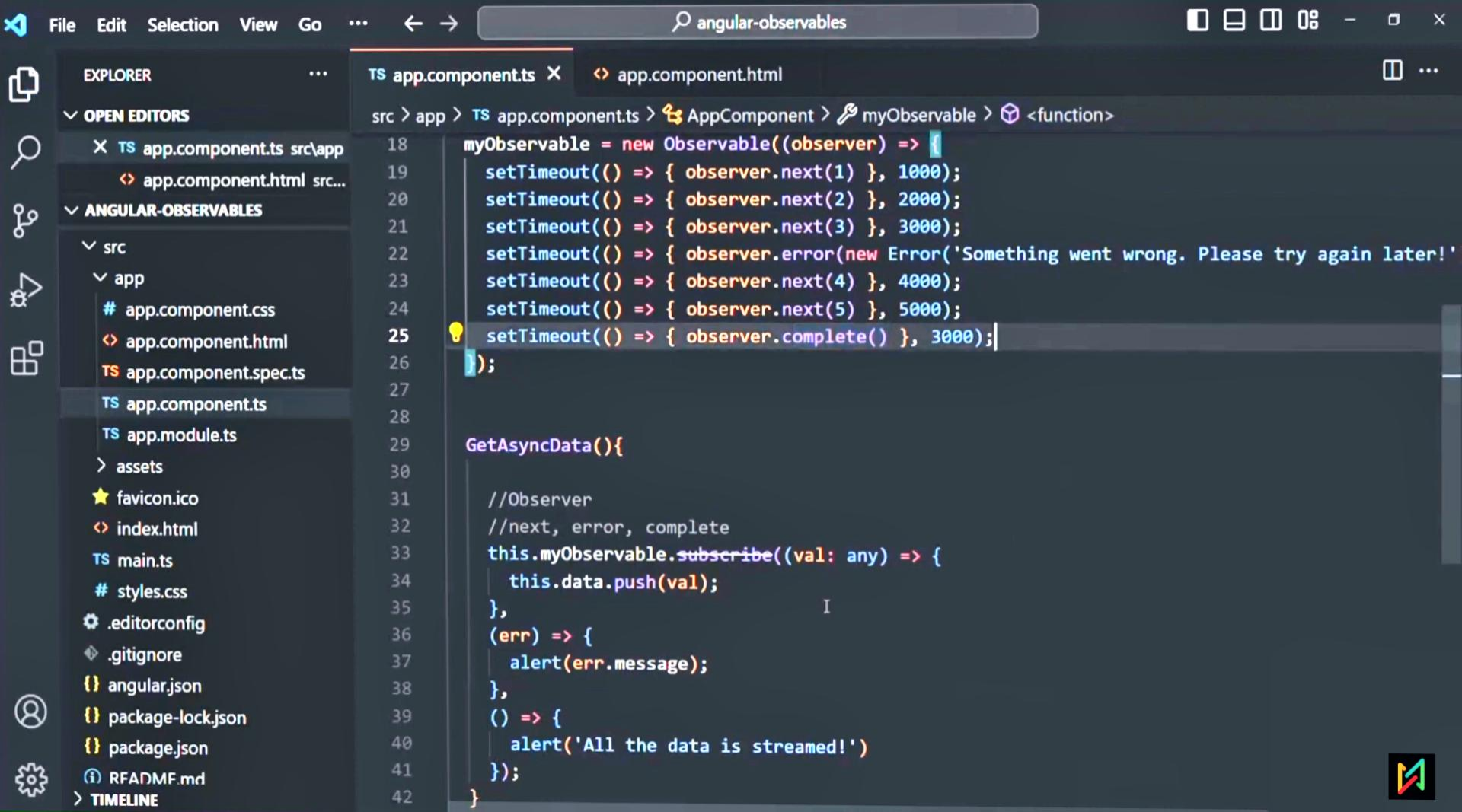
```
37     on message):
```

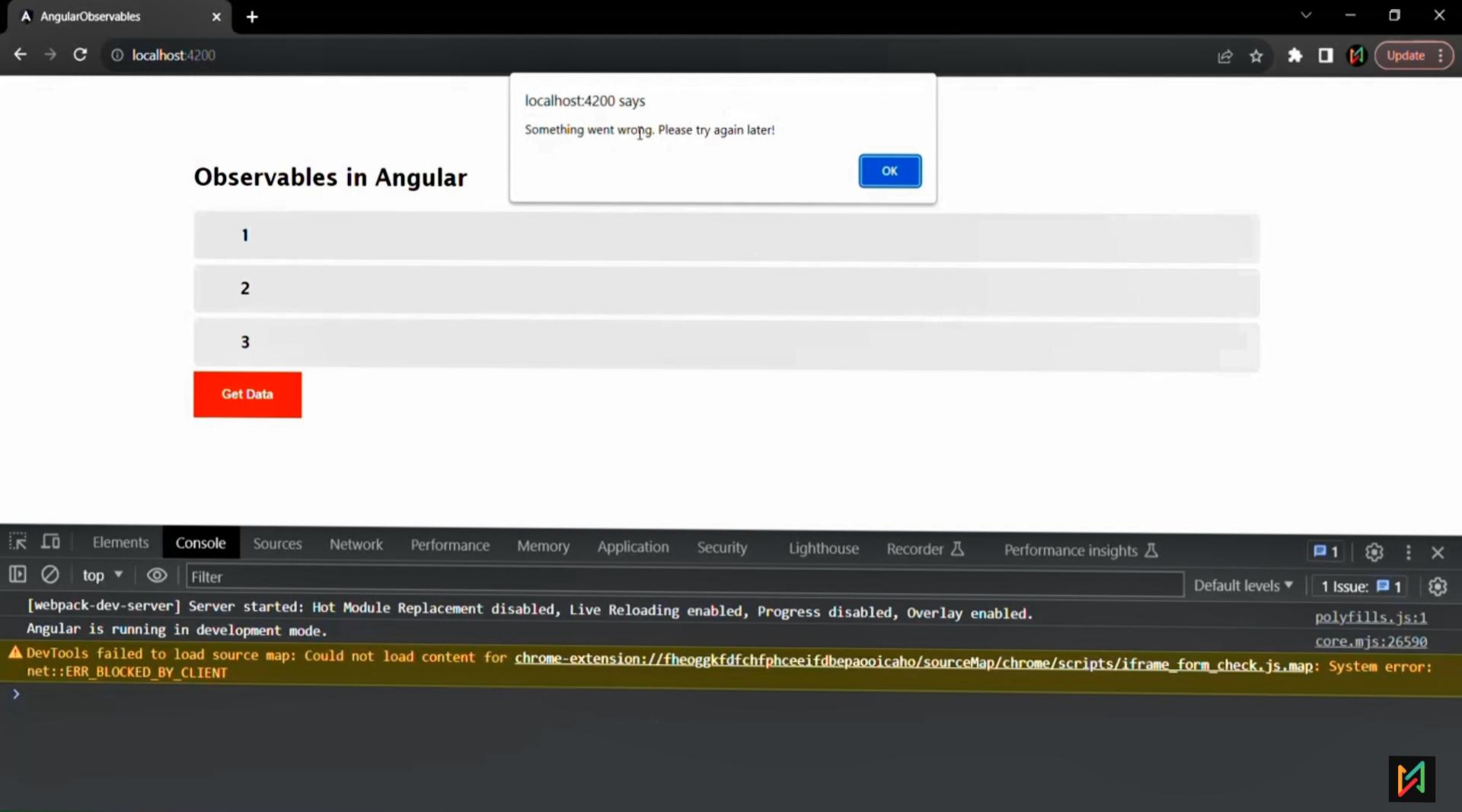
```
37         if (t.message),
```

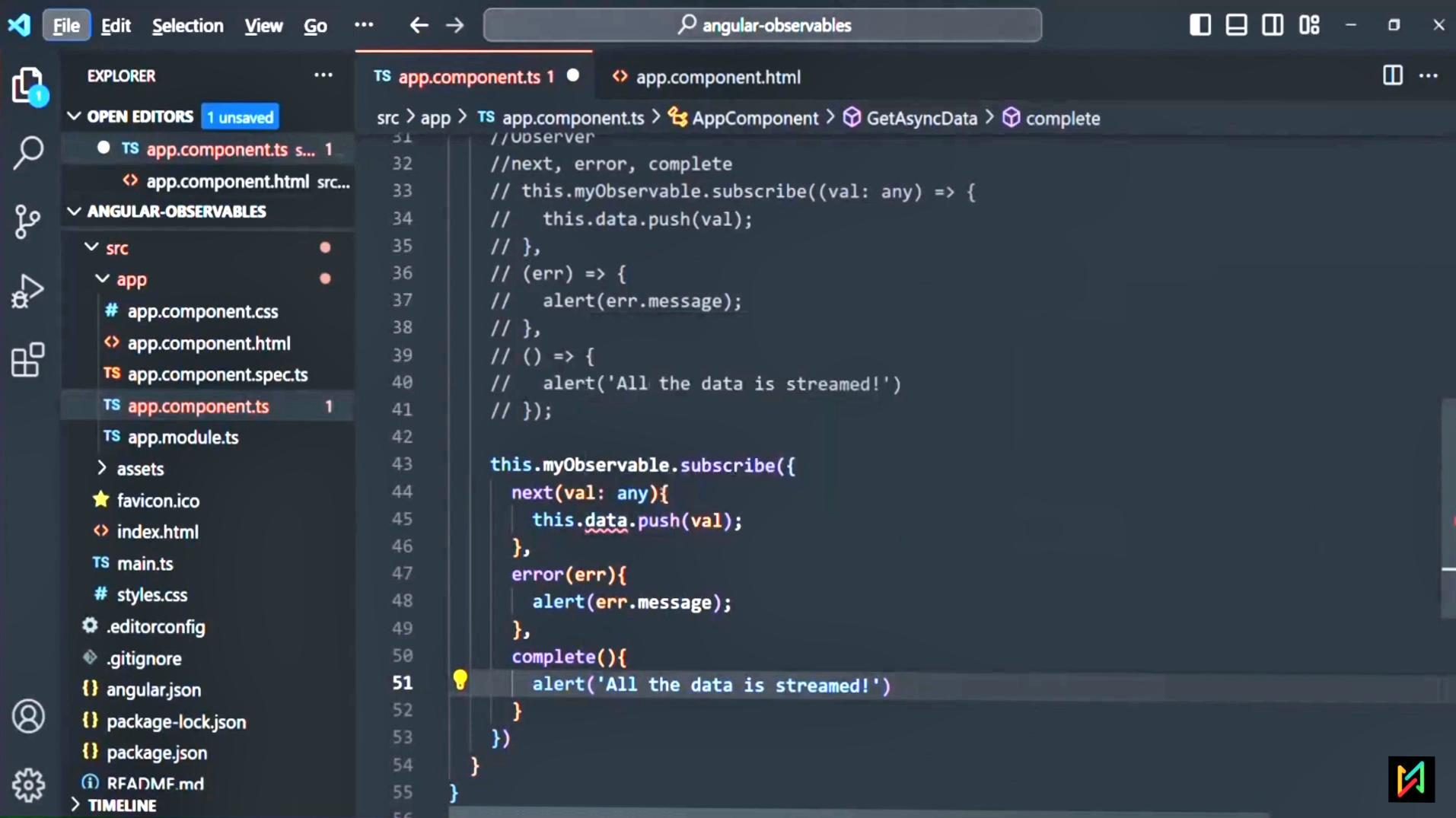
```
40 All the data is streamed!')
```

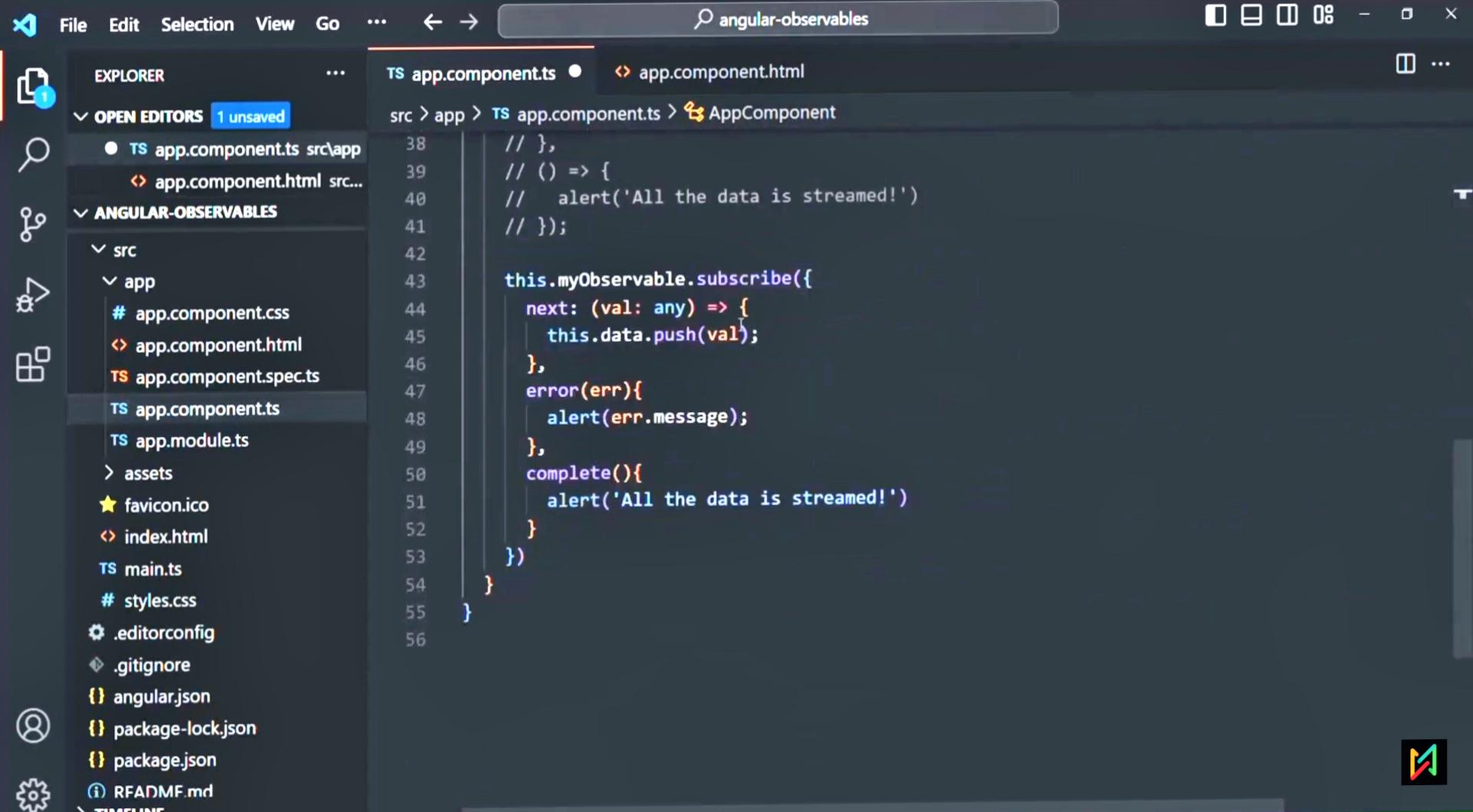
41









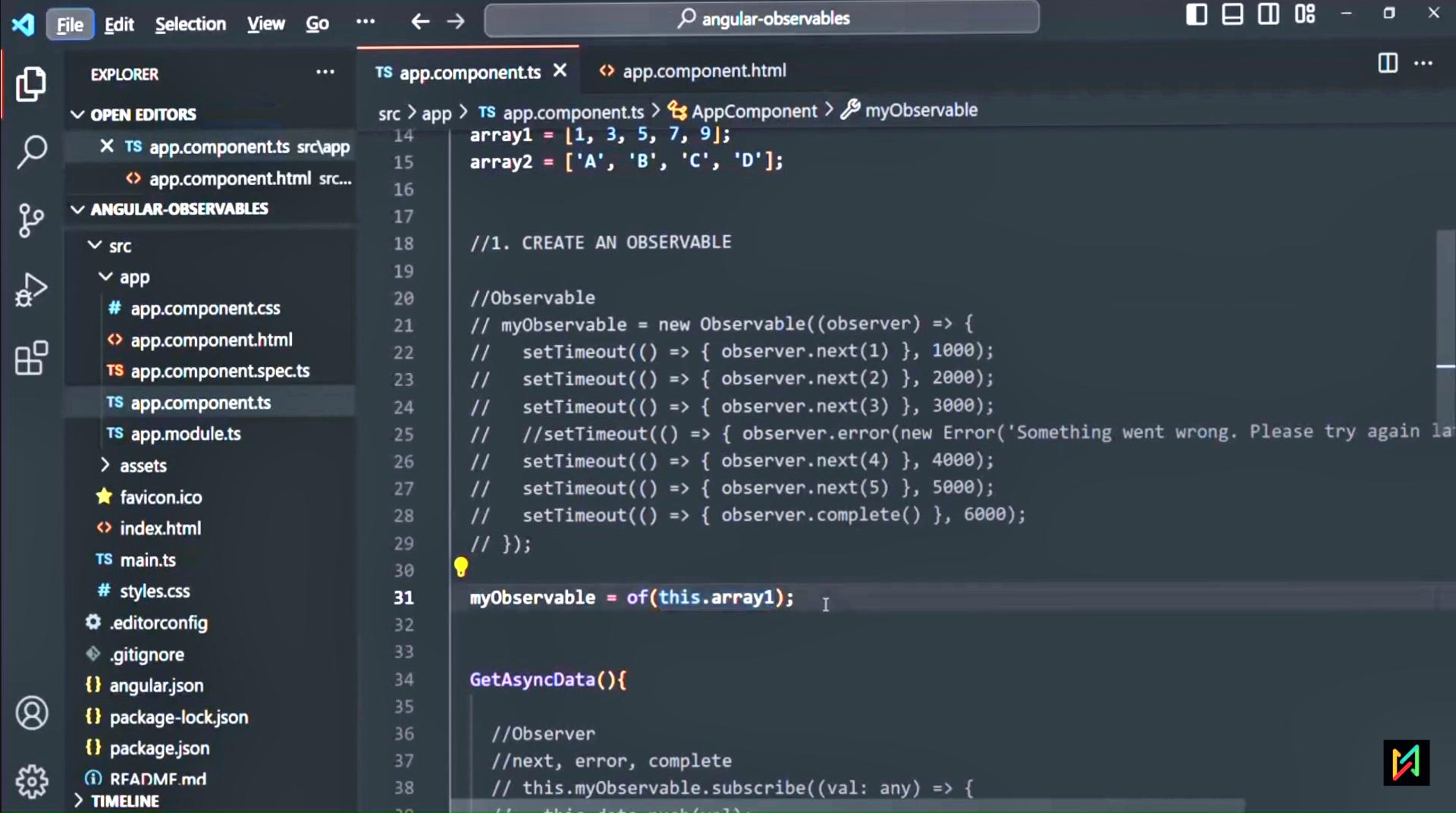


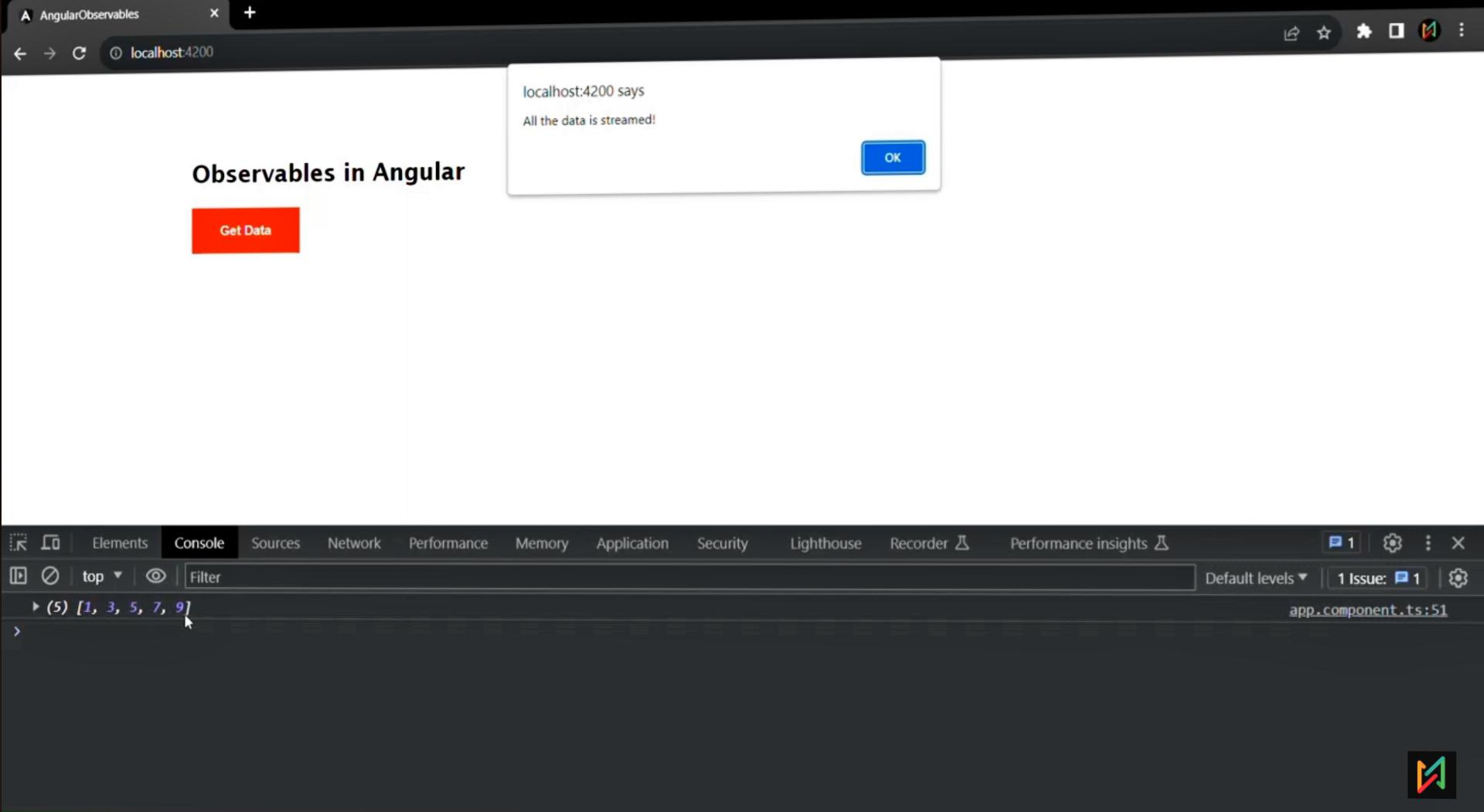
The of Operator

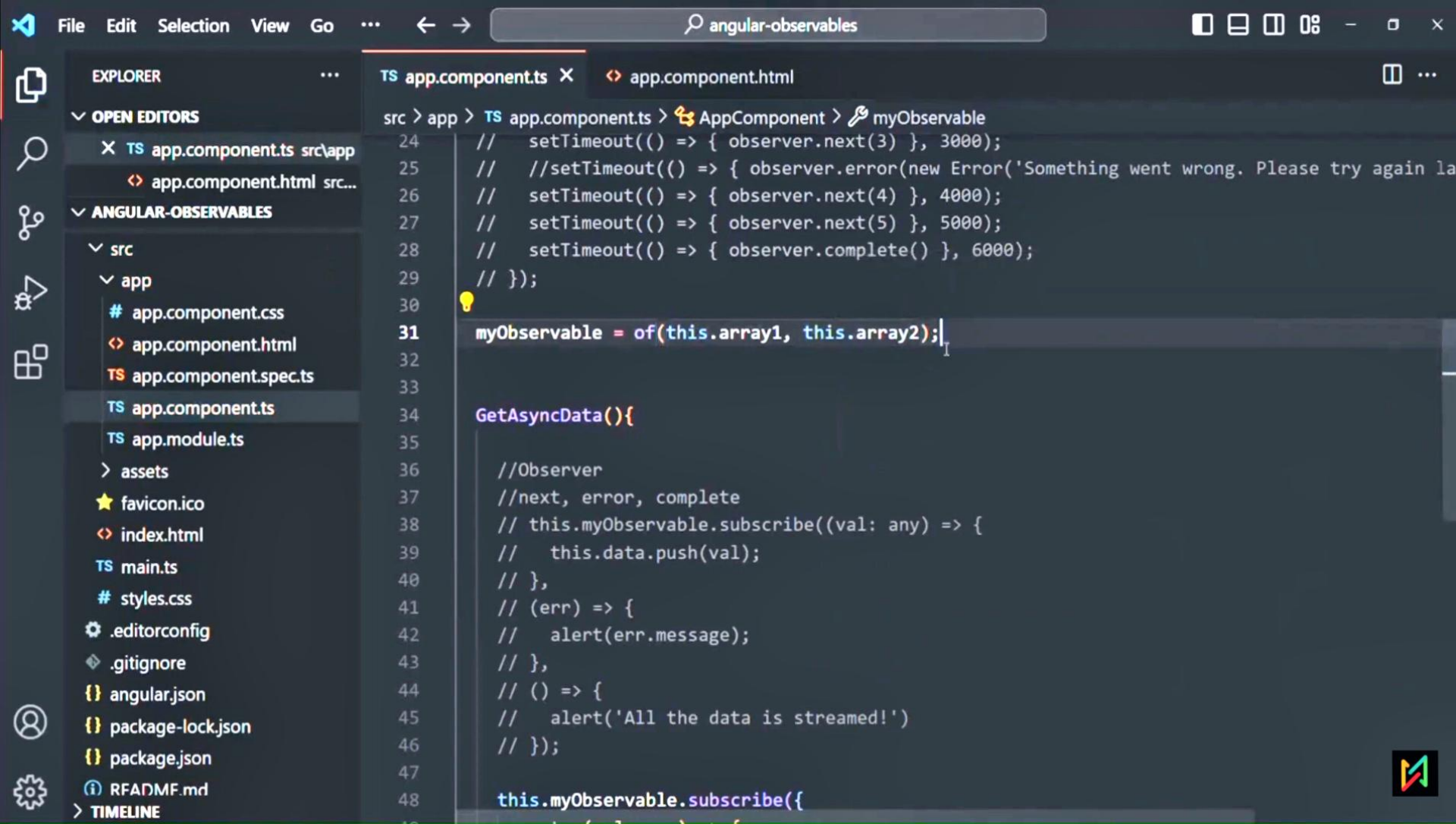
The `of` operator creates an observable from the arguments that we pass into it. You can pass any number of arguments to `of` operator.

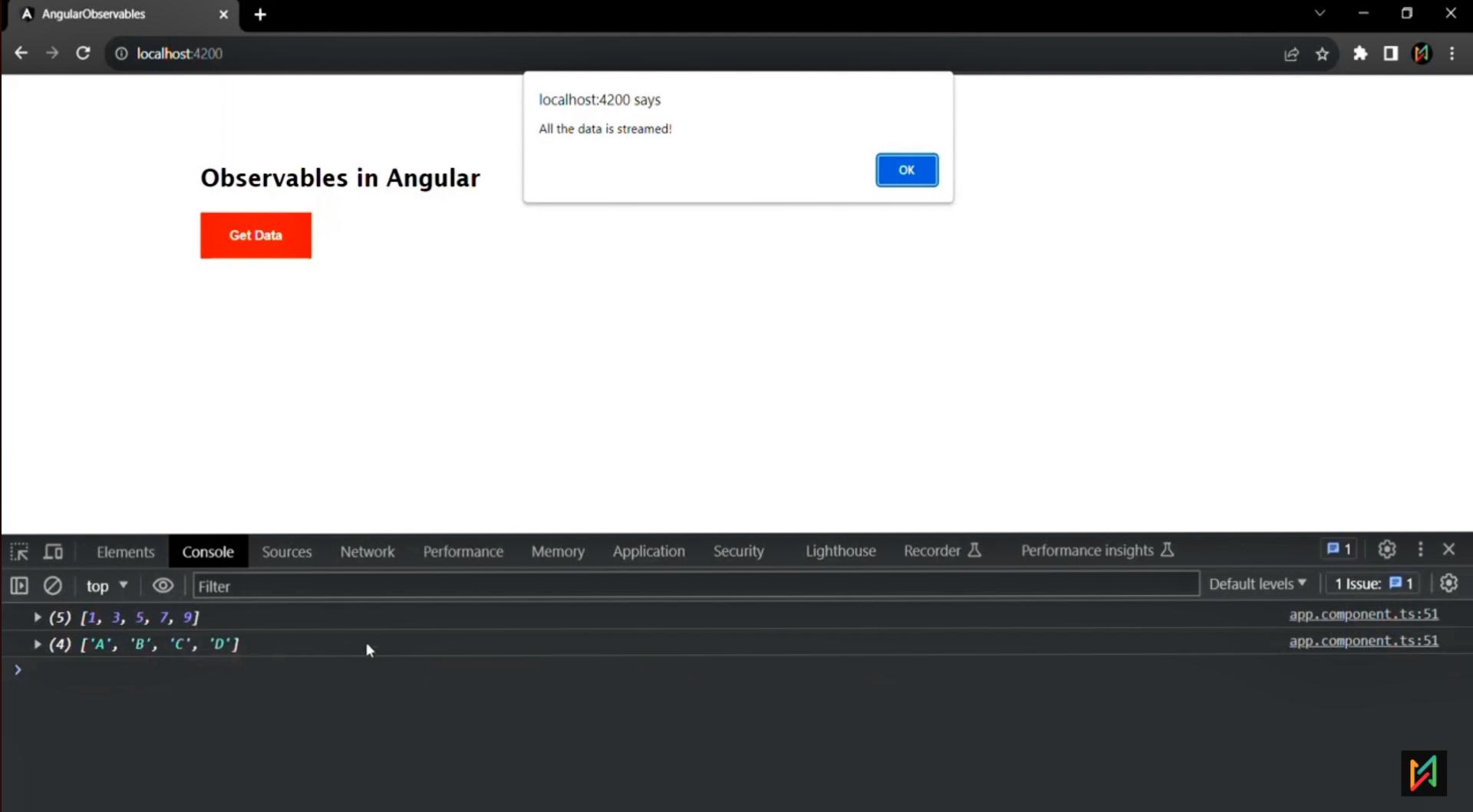
Each argument is emitted separately one after the other. It send the complete signal at the end.









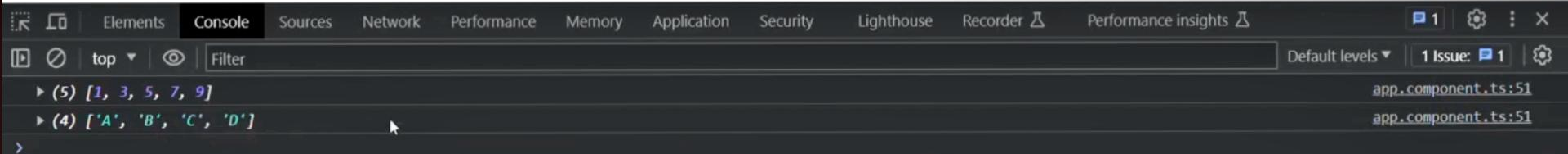


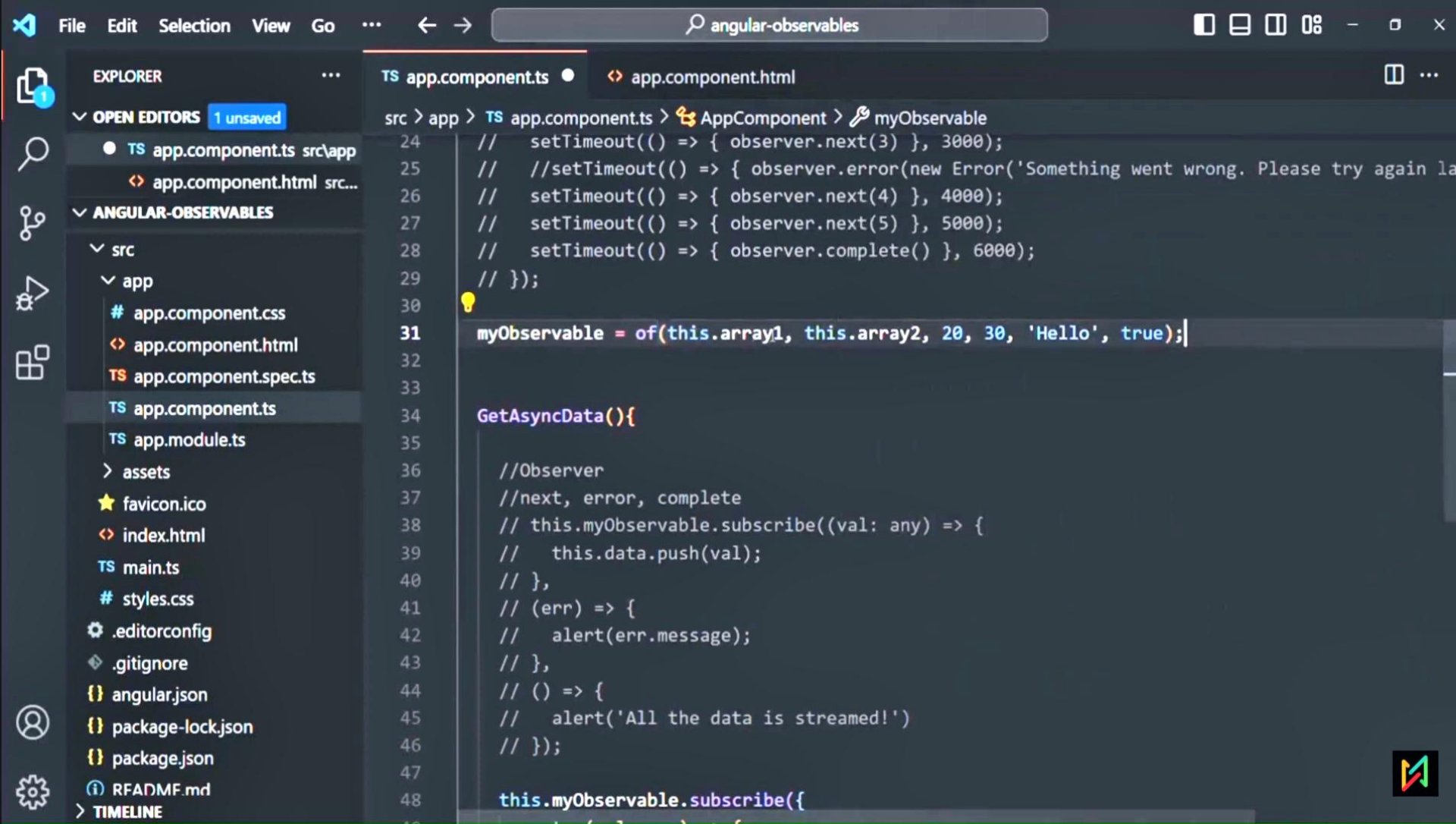
Observables in Angular

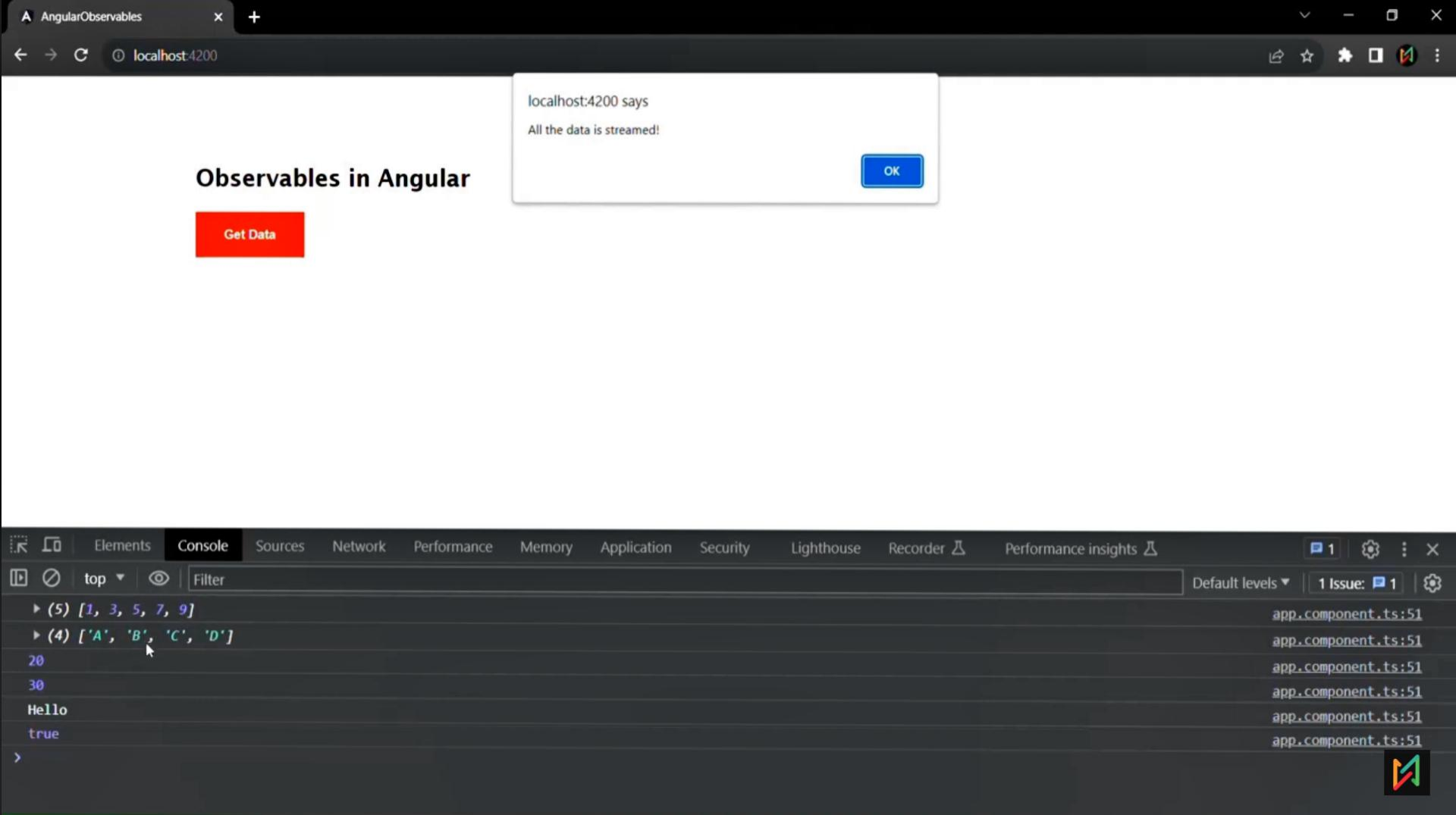
Get Data

localhost:4200 says
All the data is streamed!

OK







Observables in Angular

Get Data

localhost:4200 says
All the data is streamed!

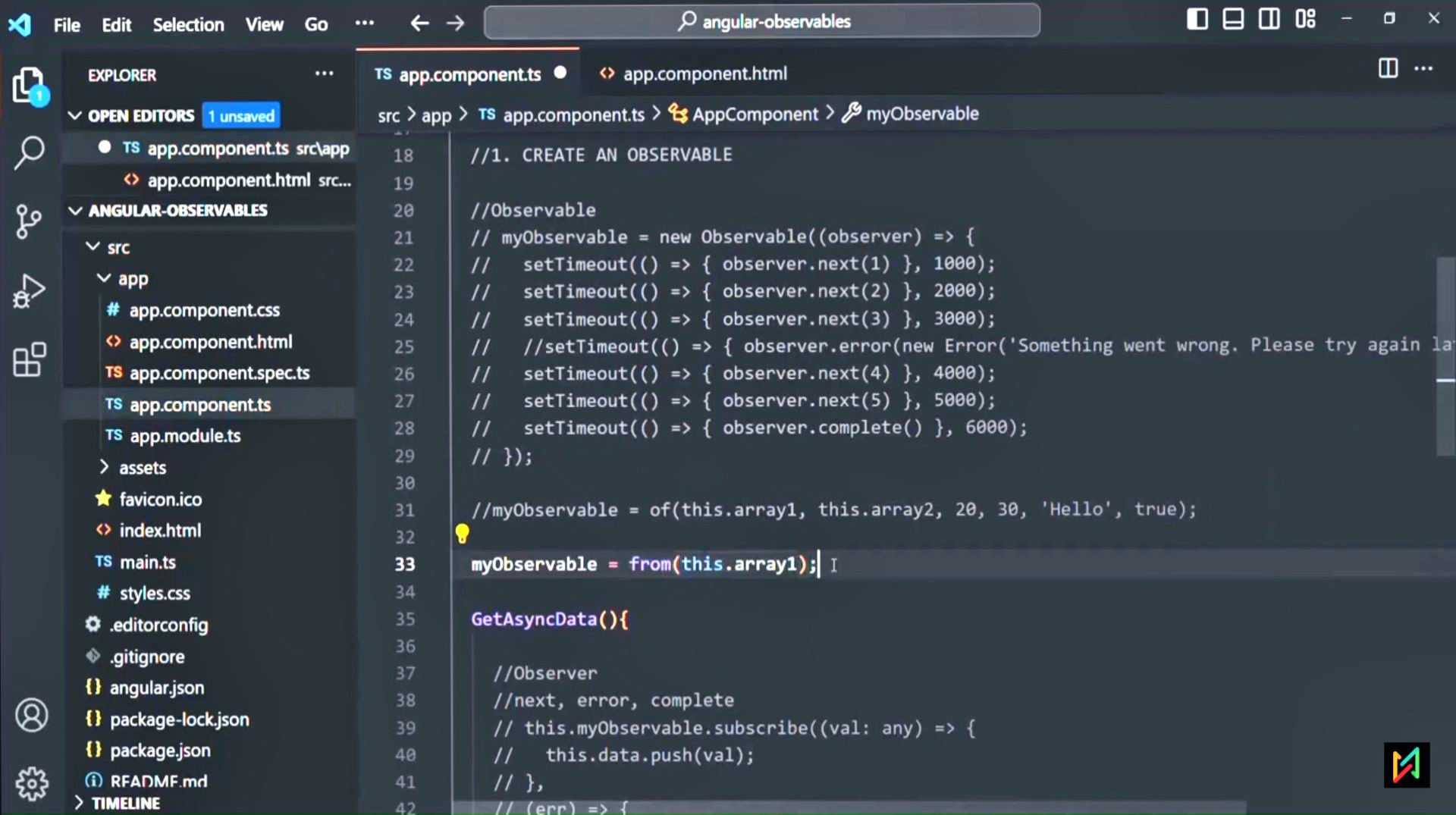
OK



The from Operator

The from operator takes a single argument which can be iterated over and converts it into an observable.





File Edit Selection View Go ...

angular-observables

0% -

EXPLORER

OPEN EDITORS 1 unsaved

TS app.component.ts src\app

app.component.html src...

ANGULAR-OBSERVABLES

src

app

app.component.css

app.component.html

TS app.component.spec.ts

TS app.component.ts

TS app.module.ts

assets

★ favicon.ico

index.html

TS main.ts

styles.css

.editorconfig

.gitignore

angular.json

package-lock.json

package.json

RFADMF.md

TIMELINE

TS app.component.ts app.component.html

src > app > TS app.component.ts > AppComponent > myObservable

18 //1. CREATE AN OBSERVABLE

19

20 //Observable

21 // myObservable = new Observable((observer) => {

22 // setTimeout(() => { observer.next(1) }, 1000);

23 // setTimeout(() => { observer.next(2) }, 2000);

24 // setTimeout(() => { observer.next(3) }, 3000);

25 // //setTimeout(() => { observer.error(new Error('Something went wrong. Please try again la

26 // setTimeout(() => { observer.next(4) }, 4000);

27 // setTimeout(() => { observer.next(5) }, 5000);

28 // setTimeout(() => { observer.complete() }, 6000);

29 // });

30

31 //myObservable = of(this.array1, this.array2, 20, 30, 'Hello', true);

32

33 myObservable = from(this.array1);

34

35 GetAsyncData(){

36

37 //Observer

38 //next, error, complete

39 // this.myObservable.subscribe((val: any) => {

40 // this.data.push(val);

41 // },

42 // (err) => {

AngularObservables

localhost:4200

Observables in Angular

1

3

5

7

9

Get Data

ElementsConsoleSourcesNetworkPerformanceMemoryApplicationSecurityLighthouseRecorderPerformance insights

topFilter

1 Issue: 11 hidden

1

3

5

7

9

>

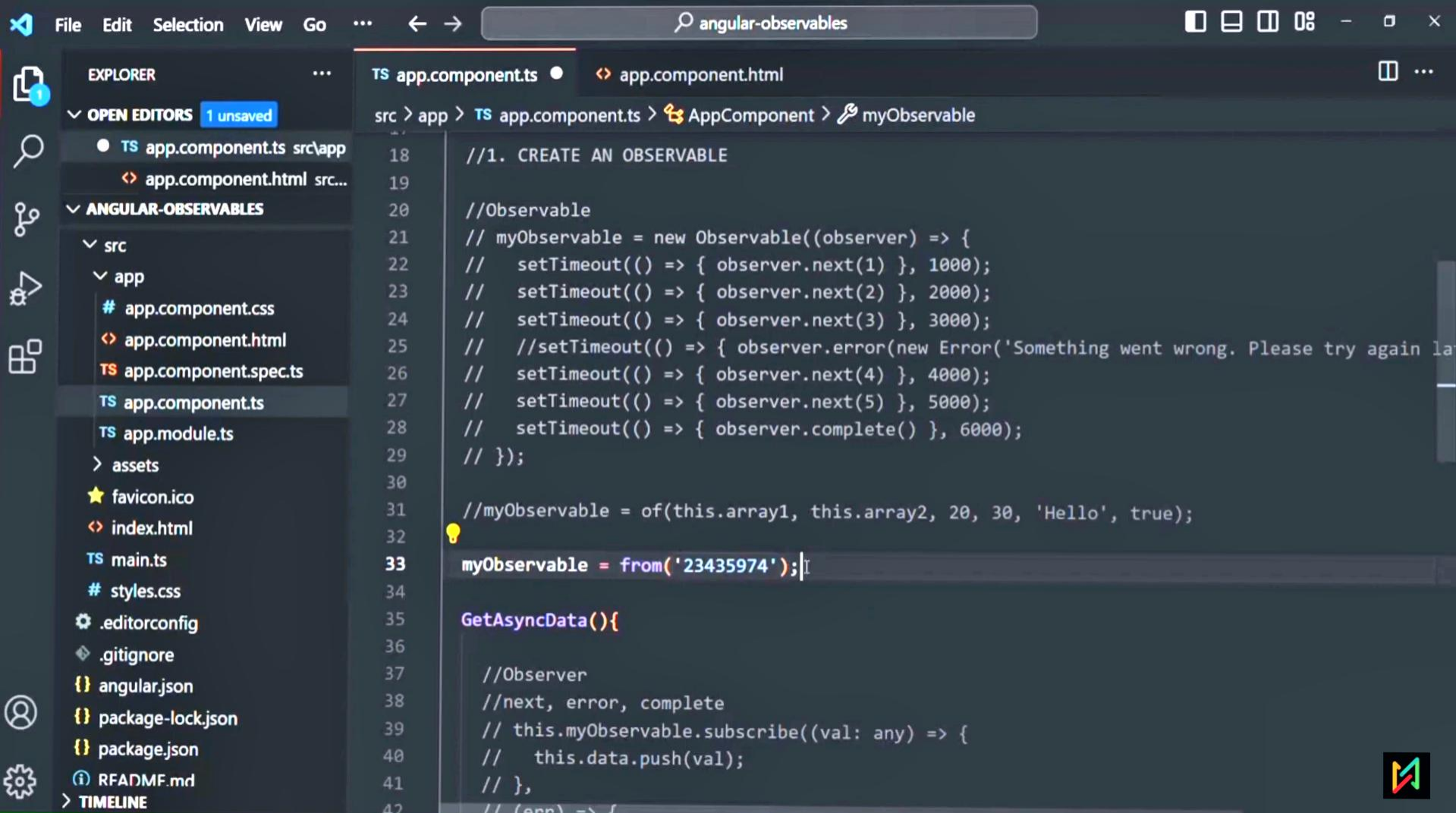
app.component.ts:52

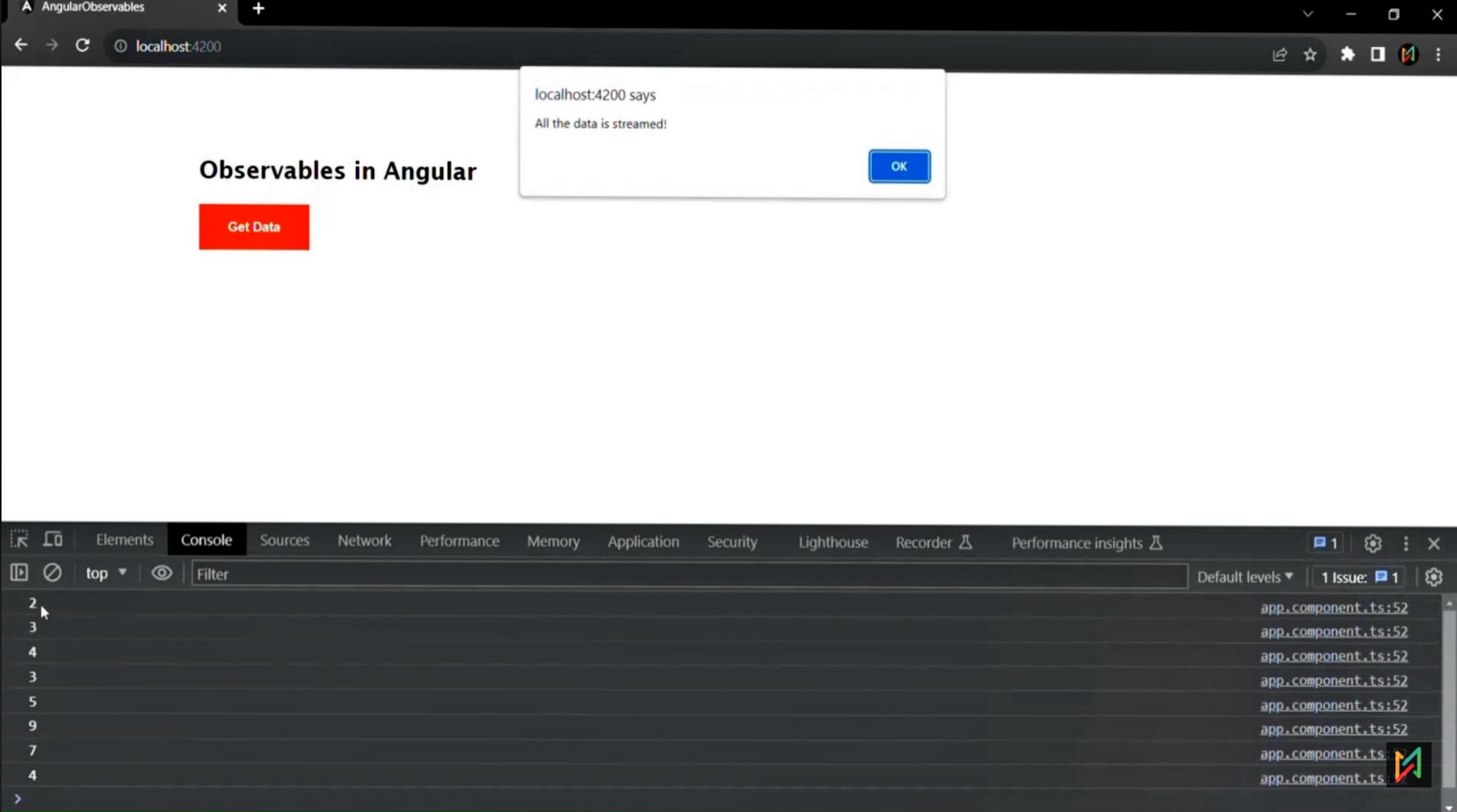
app.component.ts:52

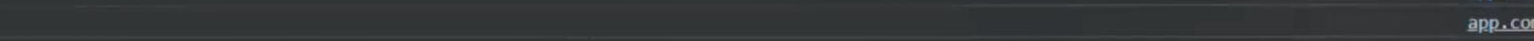
app.component.ts:52

app.component.ts:52

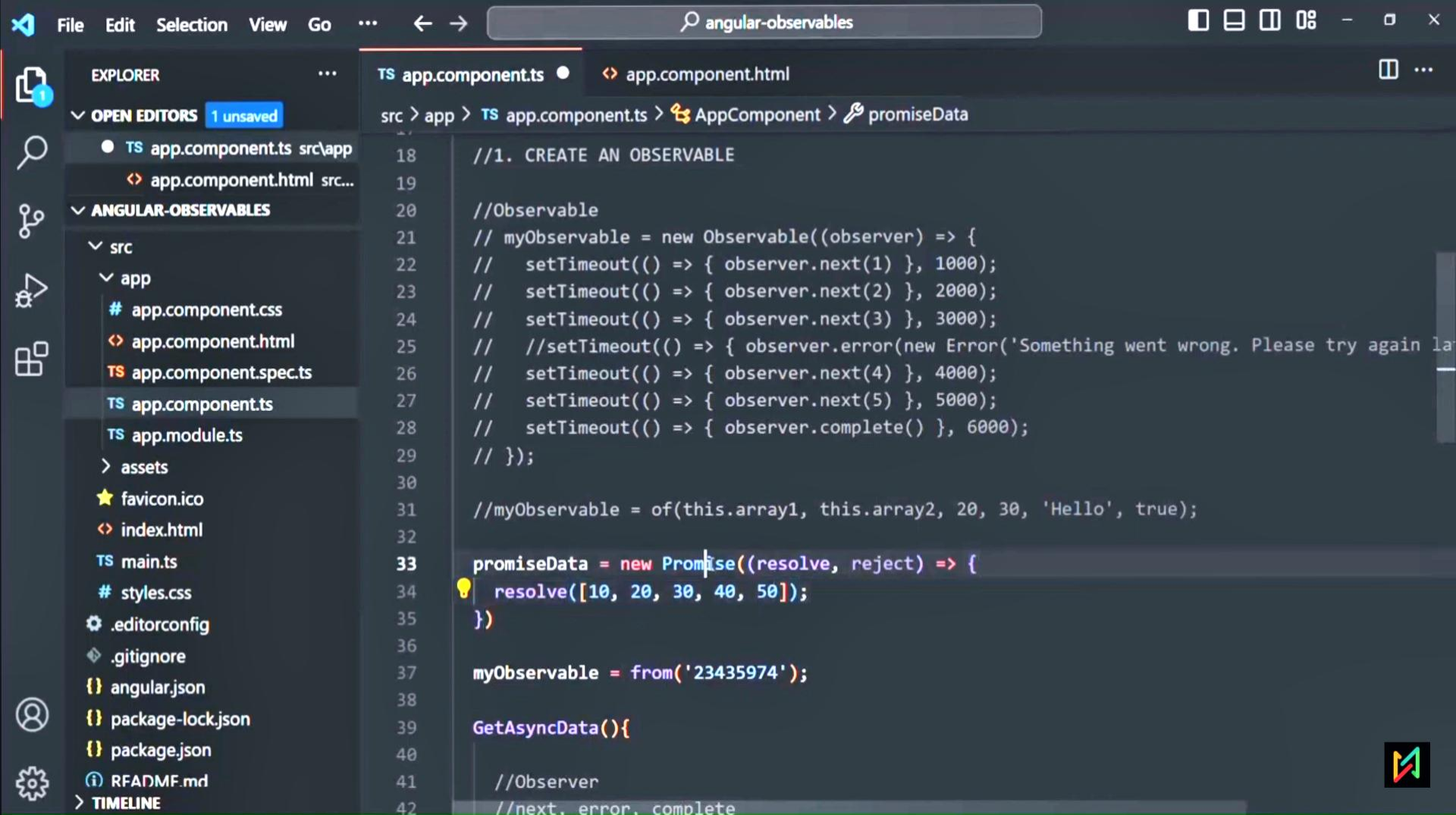
app.component.ts:52







The screenshot shows the Chrome DevTools Console with the 'Filter' bar at the top. Below the filter bar, there is a list of error messages. The messages are truncated and show 'app.component.ts:52' as the source. The console has a filter bar at the top and a list of messages below.





EXPLORER

OPEN EDITORS

TS app.component.ts src\app
app.component.html src...

ANGULAR-OBSERVABLES

src

app

app.component.css
app.component.html
TS app.component.spec.ts
TS app.component.ts

app.module.ts

> assets

★ favicon.ico
index.html

TS main.ts

styles.css

.editorconfig

.gitignore

{ } angular.json

{ } package-lock.json

{ } package.json

i RFADMF.md

> TIMELINE

TS app.component.ts X app.component.html

src > app > TS app.component.ts > AppComponent > myObservable

```
18 //1. CREATE AN OBSERVABLE
19
20 //Observable
21 // myObservable = new Observable((observer) => {
22 //   setTimeout(() => { observer.next(1) }, 1000);
23 //   setTimeout(() => { observer.next(2) }, 2000);
24 //   setTimeout(() => { observer.next(3) }, 3000);
25 //   //setTimeout(() => { observer.error(new Error('Something went wrong. Please try again later')); }, 4000);
26 //   setTimeout(() => { observer.next(4) }, 4000);
27 //   setTimeout(() => { observer.next(5) }, 5000);
28 //   setTimeout(() => { observer.complete() }, 6000);
29 // });
30
31 //myObservable = of(this.array1, this.array2, 20, 30, 'Hello', true);
32
33 promiseData = new Promise((resolve, reject) => {
34   resolve([10, 20, 30, 40, 50]);
35 });
36
37 myObservable = from(this.promiseData);
38
39 GetAsyncData(){
40
41   //Observer
42   //next, error, complete
```



